

ComicCraft

AI Comic Story Creator Using Gemini Models

Project Development Phases

This document presents the major phases followed in the development of ComicCraft, an AI-powered application that uses Google Gemini models to help users create comic stories.

1. Brainstorming & Ideation Phase

Problem Identified

Many users have creative story ideas but find it difficult to develop them into complete comic stories. Creating characters, dialogues, scenes, and a proper storyline manually can be time-consuming and challenging.

Idea

Build an AI-powered comic story creator using the Google Gemini API that transforms a simple user prompt into a structured comic story with a title, characters, plot, scenes, dialogues, and panel descriptions.

Why This Idea

- Combines story generation, character development, dialogue generation, and scene planning in one application.
- Uses generative AI to create personalized and creative stories based on user input.
- Makes comic creation easier for beginners and users without advanced storytelling skills.
- Allows users to modify or regenerate parts of a story.
- Provides a practical and manageable use case for Gemini models within the project timeline.

Project Name Chosen

ComicCraft – AI Comic Story Creator Using Gemini Models

2. Requirement Analysis Phase

Objective

Identify the functional and technical requirements needed to build the ComicCraft application.

Functional Requirements

- User can enter a story idea or prompt.
- User can select a genre such as adventure, comedy, fantasy, mystery, or science fiction.
- System can generate a story title, characters, plot, scenes, and dialogues.
- System can divide the story into comic panels.
- User can edit or regenerate generated content.
- Generated content can be displayed in a clear comic-script format.

- Application can handle invalid input and API errors.

Non-Functional Requirements

- User-friendly interface.
- Reasonable response time.
- Reliable Gemini API communication.
- Secure handling of API keys.
- Consistent story and character information.
- Maintainable and scalable application structure.

Software Requirements

Google Gemini API, HTML/CSS/JavaScript or React, a suitable backend such as Python or Node.js, a modern web browser, and optional database storage such as Firebase or MongoDB.

Hardware Requirements

Computer or laptop with at least 4 GB RAM, sufficient storage, and a stable internet connection.

3. Project Design Phase

System Design

Design the overall architecture and flow of ComicCraft before implementation.

Main Components

- User Interface – accepts prompts and displays generated comic content.
- Prompt Processing Module – prepares user input for the Gemini model.
- Gemini AI Module – sends requests to the Gemini API and receives generated content.
- Story Management Module – organizes the title, plot, characters, scenes, and dialogues.
- Panel Generation Module – structures the story into individual comic panels.
- Storage Module – optionally saves user stories and generated content.

Basic Data Flow

User Prompt → Input Processing → Gemini API → Story Generation → Panel Structuring → Comic Output

Design Considerations

- Simple navigation and readable output.
- Clear separation between frontend, backend, and AI services.
- Validation of user input.
- Secure storage of API credentials.
- Error handling for network and API failures.

4. Project Planning Phase

Objective

Create a practical schedule and divide the project into manageable tasks.

Major Tasks

- Finalize project idea and objectives.
- Collect and analyze requirements.
- Design UI and system architecture.
- Set up the development environment and Gemini API.
- Develop story-generation functionality.
- Develop character, dialogue, scene, and panel modules.
- Integrate and test all modules.
- Prepare documentation and presentation.
- Conduct the final demonstration.

Suggested Timeline

Planning can be divided into weekly milestones, with early weeks focused on requirements and design, middle weeks on development and integration, and final weeks on testing, documentation, and demonstration.

5. Project Development Phase

Objective

Implement the ComicCraft system according to the approved requirements and design.

Development Steps

- Create the frontend interface for entering story prompts.
- Connect the application to the backend.
- Configure Gemini API access securely.
- Create prompts for story generation.
- Generate titles, characters, plots, scenes, and dialogues.
- Convert the generated story into panel-by-panel content.
- Add editing and regeneration features.
- Add loading states and error messages.
- Integrate optional database storage.
- Connect all modules into one working application.

Story Generation Example

A user can enter: "Create a funny adventure story about three college students who discover a mysterious room." ComicCraft sends the requirements to Gemini and returns a structured story that can be converted into comic panels.

6. Project Testing Phase

Objective

Verify that ComicCraft works correctly, produces useful outputs, and handles unexpected situations.

Testing Types

- Unit Testing – test individual functions and modules.
- Integration Testing – verify communication between the frontend, backend, Gemini API, and storage.
- Functional Testing – check story generation, editing, regeneration, and panel creation.
- Usability Testing – check whether users can easily understand and operate the interface.
- Performance Testing – observe response time and application behavior.
- Error Testing – test invalid prompts, empty input, network failures, and API errors.

Sample Test Cases

- Submit a valid story prompt → system should generate a story.
- Submit an empty prompt → system should request valid input.
- Select a genre → generated story should follow the selected genre.
- Request regeneration → system should provide revised content.
- Simulate API failure → application should show a suitable error message.

7. Project Documentation Phase

Objective

Record the development process, technical details, results, and learning outcomes of the project.

Documentation Contents

- Project title and abstract.
- Problem statement and objectives.
- Brainstorming and ideation.
- Requirement analysis.
- System architecture and design.
- Technology stack.
- Module descriptions.
- Implementation details.
- Testing and test cases.
- Results and screenshots.
- Advantages and limitations.
- Future enhancements.
- Conclusion and references.

Importance

Good documentation makes the project easier to understand, evaluate, maintain, and demonstrate. It also provides a clear record of the decisions made during development.

8. Project Demonstration Phase

Objective

Present the completed ComicCraft application and demonstrate its main features.

Demonstration Flow

- Introduce the project and problem statement.
- Explain the role of Gemini models.
- Open the ComicCraft application.
- Enter a sample story prompt.
- Select the required genre and story options.
- Generate the story.
- Show generated characters, scenes, dialogues, and panels.
- Demonstrate editing or regeneration.
- Explain testing results and project outcomes.
- Discuss limitations and possible future improvements.

Expected Demonstration Output

The final demonstration should show how a simple idea entered by the user is transformed by ComicCraft into a structured comic story ready for further visual comic creation.

Overall Project Workflow

Phase	Main Outcome
1. Brainstorming & Ideation	Project idea, problem, and purpose finalized
2. Requirement Analysis	Functional and non-functional requirements identified
3. Project Design	Architecture, modules, data flow, and interface planned
4. Project Planning	Tasks, milestones, and development schedule prepared
5. Project Development	ComicCraft application implemented and integrated
6. Project Testing	Features and system behavior verified
7. Project Documentation	Project report and technical records prepared
8. Project Demonstration	Completed application presented and demonstrated

Conclusion: ComicCraft follows a systematic development process from idea generation and requirement analysis to design, implementation, testing, documentation, and final demonstration. The integration of Gemini models provides the core AI capability for transforming user ideas into structured comic stories.